

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 222 — Past Questions & Materials
by Renance

Course: MTH 222
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200

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Paper 1 of 3 — MTH 222 Linear Algebra Second Semester 2020 2021

Level 200 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE

SECOND SEMESTER EXAMINATION 2020/2021

Course Code/Title: MTH 222 — Linear Algebra IJ

Instructions: Attempt any 4 questions. Time Allowed: 2 ; Hours.

Question 1 (17.5 Marks)

$\begin{bmatrix} 2 & -4 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix}$

(a) Find the determinant of the matrix A using reduction of order given that $A = \begin{bmatrix} 1 & -1 & 2 \\ -1 & -1 & 2 \end{bmatrix}$

(8.5 marks).

(b) Use Cramer's rule to solve the following system of equations. (9 marks)

$-7x_4 - 6x_3 - 12x_2 = 24$

$5x_1 + 5x_2 + 7x_3 = 24$

$x_4 + x_1 = 5$

Question 2 (17.5 Marks)

$\begin{bmatrix} -3 & -1 & -2 \end{bmatrix}$

(a) Given that one of the eigenvalues of A is -1, find all other eigenvalues if $A = \begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ (10 marks).

(b) Determine the corresponding eigenvector of the eigenvalue $\lambda = -1$. (7.5 marks).

Question 3 (17.5Marks)

$\begin{bmatrix} 6 & 2 & 0 \end{bmatrix}$

Let $A = \begin{bmatrix} -2 & 3 & 3 \end{bmatrix}$ (a) Verify Caley-Hamilton Theorem (8 marks) (b) Find the minimal polynomial of A.

$\begin{bmatrix} 2 & -1 & 3 \end{bmatrix}$

(c) Hence or otherwise compute $A^3 - 2A^2 + A - 4I$ (7.5marks)

Question 4 (17.5Marks)

(a) Let $f(x,y) = x^T y = x_1 y_1 + x_2 y_2$ where $x = (x_1, x_2)^T, y = (y_1, y_2)^T \in \mathbb{R}^2$. Show that f is a bilinear form.

(5.5marks)

$\begin{bmatrix} 2 & -1 & 3 \end{bmatrix}$

(b) Let $A = \begin{bmatrix} 3 & 2 & -1 \end{bmatrix}$ Decompose A over $\mathbb{R}$ into triangular matrices (i) $A = LU$ (6 marks) (ii) $A = UL$ (6 marks)

$\begin{bmatrix} -1 & 2 \end{bmatrix}$

Question 5 (17.5Marks)

(a) Find a matrix P over the field $\mathbb{R}$ of real numbers such that $P^{-1}AP$ is a diagonal matrix where $A = \begin{bmatrix} 5 & 1 \\ 1 & 5 \end{bmatrix}$ (15

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marks)

(b) Hence show this diagonal form (2.5marks)

Question 6 (17.5 Marks)

(a) Explain the condition for inconsistency and two type of consistency solutions of a system of linear equations

in n unknowns based on the rank of the coefficient matrix A . (3 marks). —

(b) Use Gaussian elimination method to solve the following system of equations and express in vector form

$\mathbf{x} = \mathbf{p} + t\mathbf{v}$, where t is a real number, $\mathbf{p}$ and $\mathbf{v}$ are vectors. (12.5 marks) shear ee crge inka»

$$2x_1 - 2x_2 - x_3 + 7x_4 + x_5 = 1$$

$$4x_1 - x_2 - x_3 + 0x_4 - 5x_5 = 4$$

$$x_3 - x_4 + 3x_5 = -3$$

$$x_2 - x_3 + 3x_4 + x_5 = 0$$

(c) Find the particular solution of the system when $t = 1$. (2 marks)

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Paper 2 of 3 — MTH 222 PRATICE QUESTIONS

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MTH 222 PRATICE QUESTIONS

1. Explain the condition for inconsistency and two type of consistence solutions of a linear system of m

equations in n unknowns based on the rank of the coefficient matrix A .

2. Use Gaussian elimination method to solve the following system of equations and express your solution in the form $\vec{x} = \vec{p} + \lambda \vec{u}$, where λ is a real numbers, $\vec{p}$ and $\vec{u}$ are vectors.

$$2x_1 - 2x_2 - x_3 + 7x_4 + x_5 = 1$$

$$x_1 - x_2 - x_3 + 6x_4 - 5x_5 = 4$$

$$x_3 - x_4 + 3x_5 = -3$$

$$x_1 - x_2 + 3x_4 + x_5 = 0$$

(b) Find the particular solution of the system when $\lambda = 1$.

3. Find the determinant of the matrix A using reduction of order given that

$$\begin{bmatrix} 2 & -4 & 3 & 0 \\ 1 & -1 & -1 & 2 \\ 3 & -5 & 2 & 2 \\ 1 & -1 & 0 & 3 \end{bmatrix}$$

$$A = \begin{pmatrix} 2 & -4 & 3 & 0 \\ 1 & -1 & -1 & 2 \\ 3 & -5 & 2 & 2 \\ 1 & -1 & 0 & 3 \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & -4 & 3 & 0 \\ 1 & -1 & -1 & 2 \\ 3 & -5 & 2 & 2 \\ 1 & -1 & 0 & 3 \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & -4 & 3 & 0 \\ 1 & -1 & -1 & 2 \\ 3 & -5 & 2 & 2 \\ 1 & -1 & 0 & 3 \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & -4 & 3 & 0 \\ 1 & -1 & -1 & 2 \\ 3 & -5 & 2 & 2 \\ 1 & -1 & 0 & 3 \end{pmatrix}$$

4. Use Cramer's rule to solve the following system of equations.

$$-7x_1 - 6x_2 - 12x_3 = -33$$

$$5x_1 + 5x_2 + 7x_3 = 24$$

$$x_1 + 4x_3 = 5$$

5. Given that one of the eigenvalues of A is -1 , find all other eigenvalues.

$$A = \begin{pmatrix} -3 & -1 & -2 \\ 1 & 1 & 0 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$

(b) Determine the corresponding eigenvector of the eigenvalue $\lambda = -1$.

$$A = \begin{pmatrix} 6 & -2 & 2 \\ 2 & -1 & 3 \end{pmatrix}$$

6. Let $A = \begin{pmatrix} -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$ (a) Verify Caley-Hamilton Theorem (b) Find the minimal polynomial of

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 2 & -1 & 3 \end{pmatrix}$$

A. (c) Hence or otherwise compute $A^5 - 2A^4 + A - 4I$.

7. Let $B(x, y) = x \cdot y = x_1 y_1 + x_2 y_2$ where $x = (x_1, x_2)$, $y = (y_1, y_2) \in \mathbb{R}^2$. Show that B is a bilinear

form.

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 2 & -1 & 3 \end{pmatrix}$$

8. Let $A = \begin{bmatrix} 3 & 2 & -1 \\ -1 & 1 & 1 \end{bmatrix}$ Decompose A over $\mathbb{R}$ into triangular matrices (a) $A = LU$ (b) $A = QR$

$$A = \begin{bmatrix} 3 & 2 & -1 \\ -1 & 1 & 1 \end{bmatrix}$$

9. Find a matrix P over the field $\mathbb{R}$ of real numbers such that $P^{-1}AP$ is a diagonal matrix where $A =$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

[]. (b) Hence show this diagonal form.

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Paper 3 of 3 — MTH 222

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS
SECOND SEMESTER EXAMINATION 2022/2023

Course Code/Title: MTH 222 – Linear Algebra II
Instructions: Attempt any 4 questions. Time Allowed: 2Hours.

Question 1 (17.5 Marks)

(a) Find the determinant of the matrix A using reduction of order given that $A = \begin{pmatrix} 2 & -4 & 3 & 0 \\ 1 & -1 & -1 & 2 \\ 3 & -5 & 2 & 2 \\ 1 & -1 & 0 & 3 \end{pmatrix}$ (8.5 marks).

(b) Use Cramer's rule to solve the following system of equations. (9 marks)

$$\begin{aligned} -7x_1 - 6x_2 - 12x_3 &= -33 \\ 5x_1 + 5x_2 + 7x_3 &= 24 \\ x_1 + 4x_3 &= 5 \end{aligned}$$

Question 2 (17.5 Marks)

(a) Given that one of the eigenvalues of A is -1 , find all other eigenvalues if $A = \begin{pmatrix} -3 & -1 & -2 \\ 1 & -1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ (10 marks).

(b) Determine the corresponding eigenvector of the eigenvalue $\lambda = -1$. (7.5 marks).

Question 3 (17.5 Marks)

Let $A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$ (a) Verify Cayley-Hamilton Theorem (8 marks) (b) Find the minimal polynomial of A . (2 marks) (c) Hence or otherwise compute $A^5 - 2A^4 + A - 4I$ (7.5 marks)

Question 4 (17.5 Marks)

(a) Let $f(x, y) = x \cdot y = x_1y_1 + x_2y_2$ where $x = (x_1, x_2)$, $y = (y_1, y_2) \in \mathbb{R}^2$. Show that f is a bilinear form. (5.5 marks)

(b) Let $A = \begin{bmatrix} 2 & -1 & 3 \\ 3 & 2 & -1 \\ -1 & 1 & 1 \end{bmatrix}$ Decompose A over $\mathbb{R}$ into triangular matrices (i) $A = LU$ (6 marks) (ii) $A = UL$ (6 marks)

Question 5 (17.5 Marks)

(a) Find a matrix P over the field $\mathbb{R}$ of real numbers such that $P^{-1}AP$ is a diagonal matrix where $A = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$ (15 marks)

(b) Hence show this diagonal form (2.5 marks)

Question 6 (17.5 Marks)

(a) Explain the condition for inconsistency and two type of consistence solutions of a linear system of m equations in n unknowns based on the rank of the coefficient matrix A . (3 marks).

(b) Use Gaussian elimination method to solve the following system of equations and express your solution in the form $X = p + tv$, where t is a real numbers, p and v are vectors. (12.5 marks).

$$\begin{aligned} 2x_1 - 2x_2 - x_3 + 7x_4 + x_5 &= 1 \\ x_1 - x_2 - x_3 + 6x_4 - 5x_5 &= 4 \\ x_3 - x_4 + 3x_5 &= -3 \\ x_1 - x_2 + 3x_4 + x_5 &= 0 \end{aligned}$$

(c) Find the particular solution of the system when $t = 1$. (2 marks).

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